■ WTP-PLC 4.0: Master System Guide

Low-Cost Industrial Micro-PLC & HMI Architecture for Water Treatment Automation

Target Audience: Operators, Engineers, Field Technicians	**System Core:** ESP32-S3 (240MHz, 16MB Flash, 8MB PSRAM)
HMI Display: 1.8" Color LCD (160x128) with LVGL v8.3	**Release:** v4.0.0 — Production Verified Hardware

1. Executive Summary & Purpose (Easy Understanding)

What is WTP-PLC 4.0? It is an all-in-one, ultra-reliable industrial water controller designed to replace fragile, multi-chip Internet-of-Things (IoT) setups that rely on external clouds, phone apps, or external databases. The WTP-PLC operates **100% standalone and offline** with sub-millisecond deterministic logic, managing water transfer from raw water sumps to treatment stages.

- **Zero Cloud Latency:** No internet dropouts or subscription costs; runs entirely on local high-performance hardware.
- **Dual Industrial Control:** Automated tank float logic alongside unconditional manual manual operator override.
- **Quad Electrical Protection:** Live cutoff for Overvoltage, Undervoltage, Dry Running, and Motor Overload.
- **Smart Self-Healing Supervisor:** Recovers automatically once electrical faults normalize without requiring manual intervention.
- **Dual Monitoring Interface:** Physical 1.8" LCD at the control cabinet PLUS a browser-based SCADA Digital Twin.

2. System Operating Modes & Safety Architecture

Mode	Trigger / Condition	Pump Behavior & Safety Safeguards
AUTO MODE	Default power-on mode. Toggled by holding STOP for 2s.	• **Auto-Start:** Sump level reaches ≥ 75% • **Auto-Cutoff:** Sump drops to ≤ 20% • Respects all electrical safety interlocks.
MANUAL MODE	Operator manual override. Toggled by holding STOP for 2s.	• **START Button:** Turns pump ON unconditionally (disregards tank water level). • **STOP Button:** Turns pump OFF immediately.
E-STOP (Emergency)	Latching hardware button (GPIO 5)	• **Instant Cutoff:** Motor relay opens within sub-milliseconds. • **Safety Lockout:** System locked until operator holds **START for ≥ 3 seconds**.

Real-Time Electrical Protection Interlocks

The PLC continuously samples live AC Voltage and Current against user-calibrated safety limits (stored in non-volatile flash):

Interlock Fault	Default Range / Trip Point	HMI Banner Display	NeoPixel Color
Overvoltage	Voltage > 260 V AC	MOTOR OFF - HIGH VOLT	■ Steady Yellow
Undervoltage	Voltage < 180 V AC	MOTOR OFF - LOW VOLT	■ Steady Yellow
Dry Running	Current < 0.5 A (Cavitation)	MOTOR OFF - DRY LOAD	■ Steady Orange

Overload Jam	Current > 12.0 A (Lock Rotor)	MOTOR OFF - OVER LOAD	■ Steady Orange

3. Physical Hardware Assembly & Verified Benchmarks

The system has been completely fabricated and stress-tested on breadboard hardware with full wiring verification.

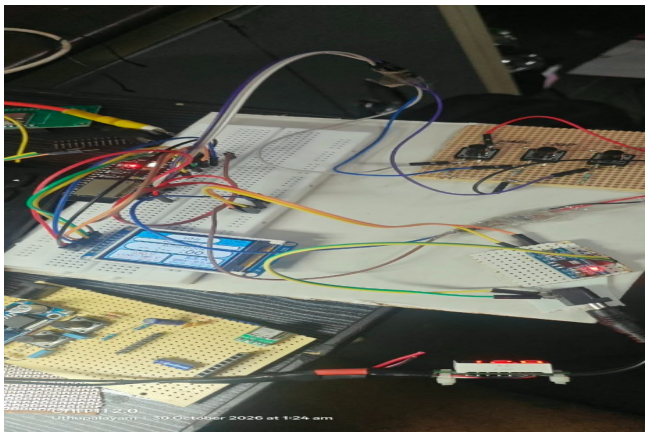


Figure 1: Physical Breadboard Setup (ESP32-S3, Buttons & LCD)



Figure 2: Live 1.8" ST7735 LVGL Industrial Interface

System Benchmark Metric	Measured Performance	Industrial Benchmark Standard
Flash Memory Usage	1.39 MB (33.3% of 4MB)	Leaves >2.6 MB free for OTA firmware upgrades
SRAM Internal Heap	119.5 KB (36.5% of 320KB)	Zero memory fragmentation; zero-heap UI updates
PLC Loop Cycle Time	**< 1 millisecond**	Sub-millisecond hard deterministic real-time response
HMI Refresh Rate	250ms Rate-Limited Snapshot	Instant forced refresh on critical state trips
WebSockets Latency	**< 5 milliseconds**	Zero perceptual lag to connected browser dashboards

4. Hardware Pinout & Circuit Connection Map

Function	ESP32-S3 Pin	Signal / Direction	Wiring Details & Purpose
MOSI / SDA	**GPIO 11**	HSPI Data (Out)	Hardware SPI data line clocked at 16 MHz
SCK / SCL	**GPIO 12**	HSPI Clock (Out)	High-speed hardware SPI clock signal
CS (Chip Select)	**GPIO 10**	Digital Output	Active LOW SPI slave select for ST7735
DC / A0	**GPIO 9**	Digital Output	LOW = Command byte, HIGH = Pixel data
RESET	**GPIO 8**	Digital Output	Active LOW hardware reset pulse at boot
LCD Backlight	**GPIO 21**	Digital Output	Active HIGH via BC547 NPN transistor
START Button	**GPIO 4**	Input (Pull-Down)	Short: Start pump; Hold 3s: Clear E-STOP
STOP Button	**GPIO 6**	Input (Pull-Down)	Short: Stop pump; Hold 2s: Toggle Mode
E-STOP Button	**GPIO 5**	Input (Pull-Down)	Instant motor trip & fault lock out
Motor Relay Out	**GPIO 7**	Digital Output	Active HIGH to 5V optocoupled contactor relay
WS2812B NeoPixel	**GPIO 48**	Single-Wire Data	Onboard RGB LED optical system status lamp

5. Embedded SCADA 4.0 Digital Twin (Browser Interface)

Served directly from the ESP32-S3's internal flash storage (http://192.168.4.1), the SCADA digital twin provides full control cabinet visibility to any laptop, tablet, or smartphone connected to the PLC's Access Point.

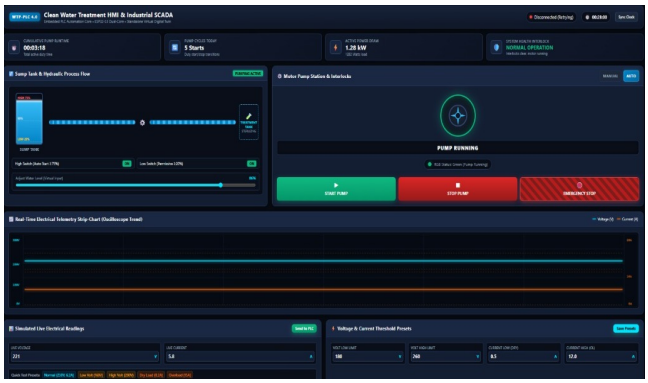


Figure 3: Full SCADA Digital Twin & P&ID; Schematic

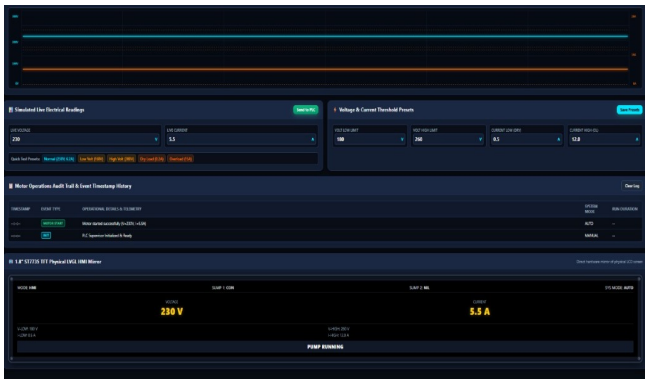


Figure 4: 60 FPS HTML5 Oscilloscope & Motor Audit Log

- **Dual-Trace 60 FPS Oscilloscope:** Hardware-accelerated HTML5 Canvas displaying real-time Voltage (Cyan) and Current (Amber) waveforms with dynamic threshold lines.
- **Motor Operations Audit Trail:** Detailed operational event logs with exact run duration computation (e.g., *"MOTOR STOP — Ran for 48s"*).
- **P&ID; Hydraulic Animation:** Sump tank with real-time level animations, float alarms, and animated flow waves passing through suction and discharge pipes.
- **Authentic TFT Mirror:** Renders an on-screen twin of the 1.8" LCD complete with industrial enclosure bezel and corner hex bolts.
- **Zero-Lag Bidirectional Sync:** Threshold adjustments and manual overrides from the browser sync across the hardware LCD in under 5ms.

6. Operator Quick-Start Guide (Step-by-Step Instructions)

Step	Operator Action	Visual Feedback & System Behavior
1	**Powering On**	NeoPixel pulses ■ Blue, display loads splash screen, then displays clock and telemetry. Default mode is AUTO.
2	**Connecting SCADA**	Connect laptop/phone to Wi-Fi AP WTP-PLC-4.0 (Password: wtp123456). Open http://192.168.4.1 in any browser.
3	**Switching Modes**	Hold the **STOP button for 2 seconds**. Banner flips between AUTO MODE and MANUAL MODE.
4	**Manual Pump Run**	In MANUAL mode, press **START** to start motor immediately. Press **STOP** to shut down.
5	**Handling E-STOP**	Pressing E-STOP cuts pump instantly and flashes ■ Red LED. To resume, clear fault and **press & hold START for ≥ 3s**.
6	**Fault Auto-Recovery**	If Voltage/Current trips out-of-range, pump halts. Once power returns within safe thresholds, pump automatically resumes.

7. Firmware Architecture & Source Code Directory

The project repository is 100% modular, documented, and completely decoupled from legacy dependencies:

- WTP_MASTER.ino: Central setup initialization, FreeRTOS scheduler, and primary loop coordinator.
- wtp_logic.cpp / .h: Demand-driven PLC state machine, electrical supervisor, and non-volatile NVS flash storage.
- buttons.cpp / .h: Non-blocking debounced button state machine with short-tap and long-hold timing logic.
- display_mgr.cpp / .h: ST7735 SPI display driver, zero-heap LVGL dirty-check refresh manager, and C-getters.
- web_server.cpp / .h: Embedded HTTP WebServer (Port 80) serving SCADA GUI + WebSockets (Port 81) telemetry streaming.
- rgb_led.cpp / .h: Non-blocking WS2812B optical signaling driver for instant diagnostic plant status codes.